

- 2 (a) Coordination in humans involves two main systems: the nervous system and the endocrine system. Paracrine cell signalling is a third way in which coordination occurs. In paracrine signalling, one cell secretes a chemical that diffuses a short distance to act upon cells that are very near to the secreting cell.

- (i) Identify **one** similarity and **one** difference between paracrine cell signalling and cell signalling that occurs as part of the endocrine system.

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- (ii) Explain why a neurotransmitter such as acetylcholine could be described as a paracrine signalling molecule.

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- (b) Human muscle cells show paracrine signalling. After muscle cells have been exposed to a substance called palmitate they:

- produce a signalling molecule
- show an increase in the expression of genes involved in a stress response (stress genes).

Scientists carried out three experiments to investigate the signalling molecule produced by muscle cells after they have been exposed to palmitate.

Experiment A

- Muscle cells were exposed to palmitate.
- The palmitate-exposed muscle cells were removed from the medium containing palmitate and then cultured in a nutrient medium for six days.
- The palmitate-exposed cells were removed from the nutrient medium and **new** muscle cells were placed in this used nutrient medium.
- The expression of stress genes in the **new** muscle cells was measured after 24 hours.
- A **control** using cells that had **not** been exposed to palmitate was also carried out.

Experiment B

Experiment **A** was repeated but the used nutrient medium for the palmitate-exposed cells and the control was boiled and then cooled before adding the new muscle cells.

Experiment C

Experiment **A** was repeated but the used nutrient medium from the palmitate-exposed cells and the control was treated to separate the lipid part. This lipid part was added to a new nutrient medium to culture the new muscle cells.

The results are shown in Fig. 2.1.

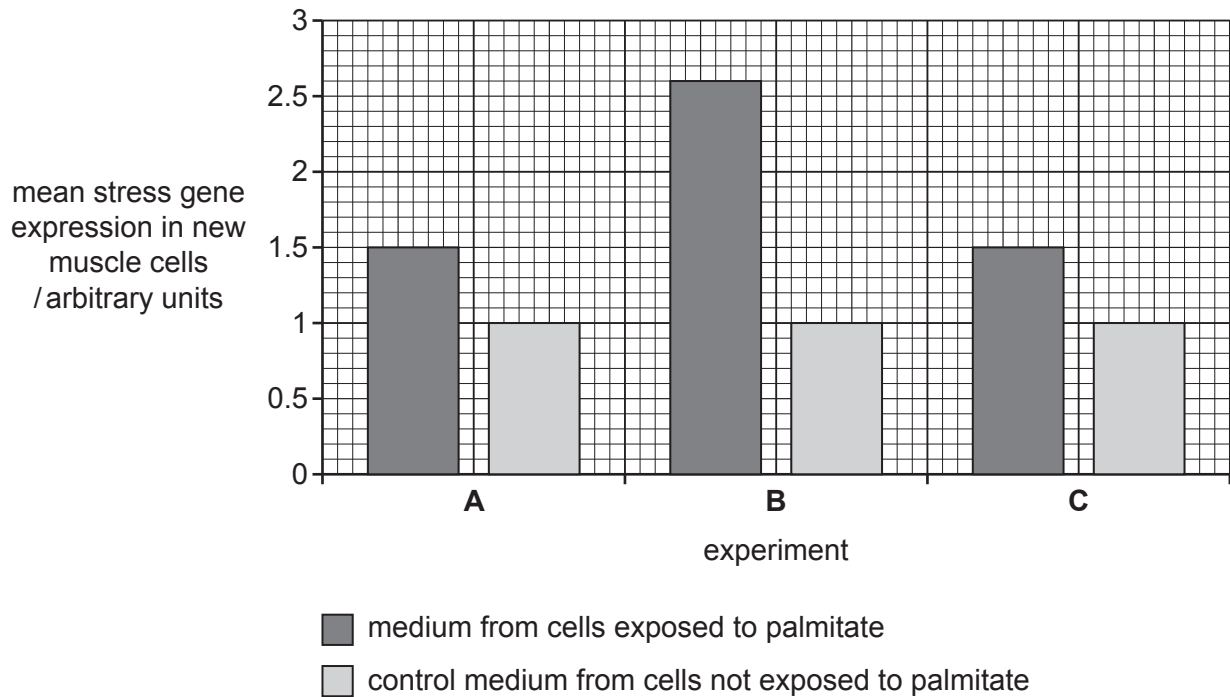


Fig. 2.1

With reference to Fig. 2.1, outline the conclusions that can be drawn about the cell signalling molecule involved in human muscle cell paracrine signalling in response to palmitate.

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- (c) The process of protein translation is inhibited in developing muscle cells when stress genes are expressed.

Suggest how developing muscle cells that express stress genes will differ in structure from normal muscle cells.

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